

2 The photograph shows a black bear.



(Source: © Cindy Hopkins/Alamy Stock Photo)

Most black bears have a diet consisting of ants and berries, and hibernate during the winter months.

One very large male black bear, called Hank, was suspected of raiding nearly 40 homes in the Lake Tahoe region of America for food. Some of these raids took place during the winter months.

- (a) Hank weighed 227 kg, which is 1.7 times more than the mean mass of a typical large black bear and 5 times more than the mass of a typical small black bear.
- (i) Calculate the difference in mass between a typical large black bear and a typical small black bear.

(2)

Answer kg

- (ii) Explain why Hank was much larger than typical black bears.

(2)

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- (b) Samples of DNA were collected from the homes that Hank was suspected of raiding.

Analysis of the DNA showed that some of the raids were by two unrelated, female black bears.

- (i) Which two procedures could have been used to analyse the DNA?

(1)

- ☐ **A** electron microscopy followed by PCR
- ☐ **B** entomology followed by electron microscopy
- ☐ **C** gel electrophoresis followed by entomology
- ☐ **D** PCR followed by gel electrophoresis

- (ii) Describe how analysis of the DNA would show that these two other bears were female, and not related to each other or to Hank.

(3)

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(Total for Question 2 = 8 marks)